

WEEK 1 TASK

Strategic Planning and Data Exploration in Logistics

Logistics Data Analysis • Week 1 Strategic Planning Report

1. Executive Summary

This project proposes a data-science workflow for an urban e-commerce logistics operation. Historical orders, inventory, delivery, vehicle and geographic data will be used to improve demand forecasting, inventory planning, delivery reliability, vehicle utilization and route efficiency.

2. Logistics Scenario

A retailer operates a distribution facility serving an urban market. Orders are picked and dispatched using a mixed fleet. Current planning relies on historical averages and manual routing. The proposed system combines descriptive analytics, prediction, clustering and constrained optimization.

3. Key KPIs

On-Time Delivery Rate = $\text{on-time deliveries} / \text{total deliveries} \times 100$. Average Delivery Lead Time = $\text{delivery time} - \text{order time}$. Inventory Stockout Rate = $\text{stockout events} / \text{demand opportunities} \times 100$. Inventory Turnover = $\text{cost of goods sold} / \text{average inventory}$. Vehicle Utilization = $\text{used capacity} / \text{available capacity} \times 100$. Cost per Delivery = $\text{total delivery cost} / \text{completed deliveries}$. Route Distance per Stop = $\text{route distance} / \text{stops}$.

4. Public Data and Research

Recommended public resources include the UCI Online Retail and Online Retail II datasets for transaction and demand analysis, and the World Bank Logistics Performance Index for logistics context. Routing methodology is grounded in Vehicle Routing Problem research, including capacity and time-window constraints. Retail data should not be presented as direct route data; route-level analysis requires delivery records.

5. Data Requirements

Orders: order_id, customer_id, product_id, order_time, quantity, price. Inventory: product, warehouse, opening stock, receipts, sales, closing stock, reorder point. Delivery: delivery_id, order_id, vehicle, dispatch time, delivery time, status, distance, service time. Geography: zone, coordinates, travel time and time windows.

6. Data Science Methods

EDA will reveal demand trends, product concentration, cancellations, zone patterns and delivery distributions. Regression/time-series models can forecast demand or delivery duration using lags, rolling averages, calendar effects and operational features. Clustering can segment customers or zones by volume, frequency, basket size and delivery characteristics. CVRP/VRPTW optimization can minimize distance, lateness, vehicle cost and fleet usage subject to capacity and time-window constraints.

7. Strategic Roadmap

Data collection → data cleaning → exploratory analysis → feature engineering → forecasting/delay-risk modeling → clustering → route optimization → KPI comparison → management recommendations. Validation will use chronological splits for forecasting, baseline comparisons, leakage checks and business-impact measures.

8. Python Illustration

Example workflow: import pandas and numpy; parse timestamps; calculate Revenue = Quantity × UnitPrice; flag cancellations; remove duplicates; aggregate daily product demand; create lag-1, lag-7 and rolling-7 features; train a RandomForestRegressor; evaluate with MAE. For segmentation, standardize customer features and apply KMeans. For routing, construct a travel-time matrix and solve a capacity/time-window constrained routing model using OR-Tools.

9. Evaluation

Forecasting: compare against a moving-average baseline using MAE/RMSE. Clustering: assess stability and operational interpretability. Routing: compare optimized versus manual routes on distance, OTD, vehicles and cost. Inventory: compare stockout and inventory levels under the proposed policy.

10. Risks and Mitigation

Public datasets may lack delivery routes: use them for demand modeling and obtain delivery data for routing. Historical demand may drift: use rolling validation and monitoring. Incomplete optimization constraints can create infeasible routes: validate capacity, service time, depot and time-window rules. Prevent leakage by using only information available at prediction time.

11. Expected Outcomes

A reproducible data model, demand baseline, forecasting prototype, operational demand segments, delivery-risk framework, routing optimization prototype and KPI scorecard. The system should support replenishment, staffing, vehicle allocation, route sequencing and service-level decisions through human-in-the-loop decision support.

12. Conclusion

The project treats logistics as a connected prediction-and-optimization problem. Starting with data quality and interpretable baselines reduces risk, while clustering and optimization translate analytical findings into operational decisions.

References

UCI Machine Learning Repository: Online Retail and Online Retail II. World Bank: Logistics Performance Index. Golden, Wang & Wasil: The Evolution of the Vehicle Routing Problem: A Survey from 2005 to 2022. Open Government Data Platform India: freight and passenger movement data. Recent research on constraint-aware express delivery route optimization.